Lecture 21

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Gödel's Completeness Theorem

- $\Phi \vdash \phi$ iff $\Phi \vDash \phi$
- What does this mean for group theory?
- What about new proof techniques?
- Once we show the equivalence between $\vdash \phi$ and $\vDash$, we can transfer properties of one to the other
  - Compactness theorem:
    (a) $\Phi \vDash \phi$ iff there is a finite $\Phi_0 \subseteq \Phi$ such that $\Phi_0 \vDash \phi$
    (b) Sat Φ iff for all finite $\Phi_0 \subseteq \Phi$, Sat Φ_0
- From the proof, we get the Löwenheim-Skolem theorem: Every satisfiable and at most countable set of formulas is satisfiable over a domain which is at most countable

Consequences of Completeness

Theorem *Every satisfiable set of formulas $\Phi \subseteq L^S$ is satisfiable over a domain of cardinality $\leq |L^S|$.*

Theorem *If $\Phi \subseteq L_0^S$ has arbitrarily large finite models (for every $i \in \omega$, Φ has a model whose domain has more than i elements), then it has an infinite model.*

Proof Let

$$\psi_2 = \exists x \exists y \; x \not\equiv y$$

$$\psi_3 = \exists x \exists y \exists z \; (x \not\equiv y) \wedge (x \not\equiv z) \wedge (y \not\equiv z)$$

$$\ldots$$

Consider $\Phi \cup \{\psi_i : i > 2\}$. Note that every finite subset has a model. By compactness, the set has a model. $\square$

Upward Löwenheim-Skolem Theorem

Theorem 10 *Let Φ be a set of formulas that is satisfiable over an infinite domain and let κ be an infinite cardinal $\geq |\Phi|$. Then Φ has a model of cardinality κ.*

- Proof Idea:
  - Add new κ new distinct constants
  - Every finite subset is SAT
  - By compactness, the resulting set of formulas is SAT
  - By Lowenheim-Skolem, there is a model of size κ

Theories

T *is a theory iff* $T \subseteq L_0^S$ *and for all* $\varphi \in L_0^S$, $T \models \varphi \Rightarrow \varphi \in T$.

$\Phi^{\models} = \{\varphi : \Phi \models \varphi\}$.

T *is a theory iff* $T = T^{\models}$

T *is* complete *iff* $\langle \forall \varphi :: \varphi \in T \vee \neg\varphi \in T \rangle$.

For a set of structures $\mathbf{K}$, $\mathrm{Th}\,\mathbf{K} = \{\varphi \in L_0^S : \forall\, \mathbf{U} \in \mathbf{K}, \mathbf{U} \models \varphi\}$

Axiomatizable Theories

Definition *Theory T is* axiomatizable *iff there exists a* decidable *set Φ of sentences s.t. $T = \Phi^{\models}$.*

Definition *Theory T is* finitely axiomatizable *iff $T = \Phi^{\models}$, where Φ is a set of sentences s.t. $|\Phi| < \omega$.*

Note that if Theory T is finitely axiomatizable $T = \{\varphi\}^{\models}$ (where φ is a conjunction of finitely many sentences).

Lemma *(a) If T is finitely axiomatizable, it is axiomatizable. (b) If T is axiomatizable, it is r.e. (c) If T is axiomatizable and complete, it is recursive.*

Non-Standard Models

- Let $N_s = \langle \omega, s, 0 \rangle$, where s is the successor function. N_s satisfies:
  - (the successor of any number differs from that number) $\langle \forall x \;\; x \neq s(x) \rangle$
  - (s is injective) $\langle \forall x, y \;\; x \neq y \;\; \Rightarrow \;\; s(x) \neq s(y) \rangle$
  - (every non-0 number has a predecessor) $\langle \forall x \;\; x \neq 0 \;\; \Rightarrow \;\; \langle \exists y \;\; x = s(y) \rangle \rangle$
- Let $\Psi = \text{Th } N_s \cup \{x \neq 0, x \neq s(0), \ldots, x \neq s^n(0), \ldots\}$
- Every finite subset of Ψ has a model, so Ψ has a model (compactness)
- By Lowenheim-Skolem, let $\mathfrak{U}$ be a countable model of Ψ
  - $\mathfrak{U}$ includes $0, s(0), \ldots, s^n(0), \ldots$, and a, a non-standard number
  - a has a successor, predecessor, and they have successors, predecessors
  - so a is part of a $\mathbb{Z}$-chain
  - hence, there is a countable model, $\mathfrak{U}$, which is *not* isomorphic to N_s
- While there is a complete axiomatization for Th N_s, once the logic is powerful enough (add +, *, <), completeness goes out the window

$0, s(0), \ldots, s^n(0), \ldots,$ $\quad \ldots, p^n(a), \ldots, p(a), \quad a, s(a), \ldots, s^n(a), \ldots$ $\quad$ $\mathbb{Z}$-chain

$p(a)$ is the predecessor of a $\quad$ (isomophic to $\mathbb{Z}$)

Gödel's 1st Incompleteness Theorem

- A set is *recursive* iff $\in$ can be decided by a Turing machine
- Assuming Con(ZF), the set $\{\phi : ZF \vdash \phi\}$ is not recursive
- More generally, for any consistent extension C of ZF:
  - $\{\phi : C \vdash \phi\}$ is not recursive
  - Intuitively clear: embed Turing machines in set theory
  - Encode halting problem! as a formula in set theory
- Theorem: If C is a recursive consistent extension of ZF, then it is incomplete, i.e., there is a formula ϕ such that $C \nvdash \phi$ and $C \nvdash \neg\phi$
- Proof Outline: If not, then for every ϕ, either $C \vdash \phi$ or $C \vdash \neg\phi$. We can now decide $C \vdash \phi$: enumerate all proofs of C. Stop when a proof for ϕ or $\neg\phi$ is found

Gödel's 2nd Incompleteness Theorem

- TM_n is the nth Turing machine
- TM is a Turing machine that given input n:
  - Searches for a proof in PA that "TM_n does not halt at n"
  - If it finds a proof, TM halts; otherwise TM does not halt
- Let TM be TM_k. What if we run TM_k at k?
- Case 1. There is a proof in PA that "TM_k does not halt at k", so:
  - TM_k halts at k
  - But then PA proves "TM_k halts at k"
  - Since Con(PA), this is impossible
- Case 2. (*)There is no proof in PA that "TM_k does not halt at k"
  - Then (#)TM_k does not halt at k
- We proved: (#) and (*), the 1st Incompleteness theorem for PA
- Also, if PA can prove Con(PA), then PA can prove (*), (#)
- Thus, PA would prove: (*) & PA proves "TM_k does not halt at k"
- Hence Inc(PA); thus PA cannot prove its own consistency

FOL Observations

- In ZF, the axiom of choice is neither provable nor refutable
- In ZFC, the continuum hypothesis is neither provable nor refutable
- By Gödel's first incompleteness theorem, no matter how we extend ZFC, there will always be sentences which are neither provable nor refutable
- There are non-standard models of $\mathbb{N}$, $\mathbb{R}$ (un/countable)
- Since any reasonable proof theory has to be decidable, and TMs can be formalized in FOL (set theory), any logic can be reduced to FOL
- Building reliable computing systems requires having programs that can reason about other programs and this means we have to really understand what a proof is so that we can program a computer to do it

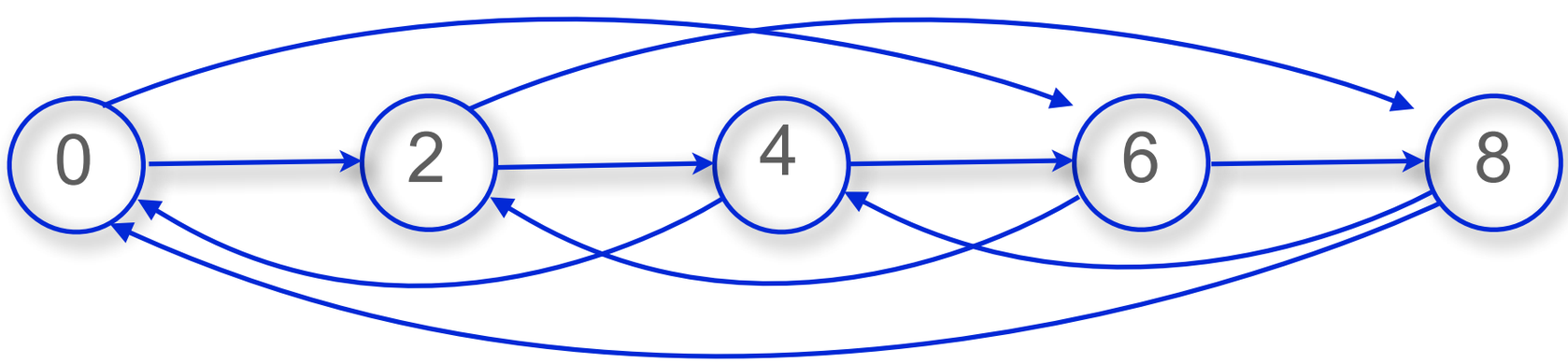
Model Checking

- Does a finite state program satisfy a temporal property?
- Search an implicit graph for errors
- Explicit state model checking
  - Start with initial states
  - Choose an unexplored state and check it
  - If error report; else generate successors & repeat
- Initially: c = 0; m = 10
- Transition relation: c := c+2 mod m || c := c+6 mod m
- Property: G(c < m-1)

finite program

property

implicit graph



Holds?

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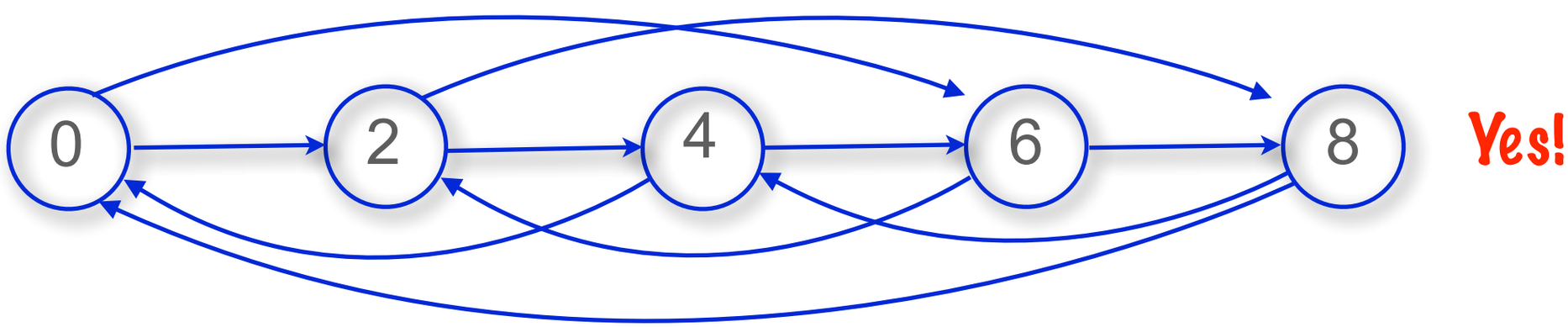

Yes!

Model Checking: Concurrency

Initially n = 0 (global, shared variable)

Processes P_i, $1 \leq i \leq m$

initially reg, counter = 0, 0 (local variables)

```
while counter < 100 {
    reg := n
    reg++
    n := reg
    counter ++ }
```

- What values can n have after all processes terminate?
- G(terminate ⇒ 100≤n≤100*m*) holds? No
- G(terminate ⇒ 2≤n≤100*m*)